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An Optimised Design of a 4-bit Carry Look-Ahead Adder Using Novel Reversible Logic Gates

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***Abstract*—Reversible logic gates are pivotal in the advancement of low-power and quantum computing technologies due to their ability to operate without generating heat from information loss. This field of study has emerged as a forefront of research, propelling innovative advancements aimed at optimizing resource efficiency. This paper presents the design and implementation of a 4-bit Carry Look-Ahead (CLA) adder using reversible logic gates, significantly enhancing arithmetic operation speed by computing carry bits in parallel. The CLA adder uses generate (G_i) and propagate (P_i) functions to determine carry bits, reducing the delay associated with traditional adders. The proposed design leverages a combination of Peres and Feynman gates to construct the Carry Propagate and Carry Generate Adder (PGA) block, producing the necessary carry and sum outputs efficiently. There is an approximate 41% improvement in the total reversible implementation cost (TRLIC) of the design. Evaluated using Vivado 2022.2 with Verilog HDL, the design demonstrates accurate and efficient performance across all 4-bit input combinations, significantly reducing propagation delay.**

***Index Terms*—Ancilla Inputs(AI), Garbage outputs(GO), Carry Look-Ahead (CLA) adder, Quantum Cost(QC) and Total Reversible Logic Implementation Cost(TRLIC).**

I. Introduction

The revolutionary potential of reversible logic gates has made them an increasingly popular domain in the Quantum computing recently. Unlike traditional logic gates, which dissipate energy in the form of heat due to irreversible operations, reversible logic gates enable computation without energy loss, adhering to the principle of zero energy dissipation as posited by Rolf Landauer in 1961. This feature makes them particularly attractive in the context of energy-efficient computing and is crucial for the development of quantum computing [1-2]. The significance of reversible logic extends beyond quantum computing into fields like nanotechnology and optical computing. In these domains, the ability to minimize heat generation and energy consumption is paramount, especially as devices continue to shrink in size. Reversible gates enable the development of circuits that produce minimal heat, thereby enhancing the longevity and reliability of nanoscale devices. Additionally, optical computing, which relies on light for data transmission and processing, benefits from the low-energy dissipation properties of reversible gates, leading to faster and more efficient optical circuits [3-4].

One of the primary advantages of reversible logic gates is their application in designing efficient arithmetic circuits, such as adders and multipliers, which are fundamental components in digital signal processing (DSP) and various computational systems. Traditional adder circuits often generate excessive garbage outputs—unnecessary intermediate data that do not contribute to the final result [5]. These outputs not only waste computational resources but also increase the overall complexity and power consumption of the circuit. Reversible logic gates address these issues by significantly reducing garbage outputs and minimizing the need for additional ancilla inputs—extra bits required to maintain reversibility. Consequently, circuits designed with reversible logic gates exhibit lower quantum costs, making them more suitable for scalable and high-performance computing applications. Some of the commonly used reversible logic gates include the Fredkin gate, the Toffoli gate, and the Peres gate, each serving as essential building blocks for constructing complex reversible circuits. These gates not only help in minimizing power consumption but also pave the way for advancements in fault-tolerant computing and cryptographic applications [6-8]. Despite the promising prospects, there are several challenges that need to be addressed to fully realize the potential of reversible logic gates. These include the development of efficient synthesis algorithms, optimizing gate count and quantum cost, and ensuring compatibility with existing technology. Ongoing research is focused on overcoming these obstacles and exploring novel applications of reversible logic [9-11].

The Carry Look-Ahead (CLA) adder is a type of digital adder used in computer arithmetic to improve speed by reducing the time required to calculate carry bits. Unlike the

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traditional ripple carry adder, where each bit must wait for the previous carry to be computed sequentially, the CLA adder precomputes the carry signals in parallel based on the inputs. This is achieved through the generation and propagation of carry signals using additional logic circuits, allowing for faster addition operations. The CLA adder is particularly effective in high-speed computing environments and is widely used in processors and digital signal processing applications where quick and efficient arithmetic operations are critical [12-15].

Here are some reversible logic gates used to design a 4-bit carry look ahead adder:

1)Feynman Gate: The Feynman gate shown in Fig. 1., also known as the Controlled NOT (CNOT) gate, is a fundamental reversible logic gate used in quantum computing and reversible computing. This gate with quantum cost of 1 not only maintains the reversibility of the operation but also plays a crucial role in entanglement and quantum error correction processes.

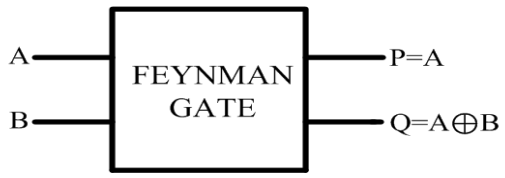


Fig. 1. Feynman Gate

2)Peres Gate: The Peres gate with quantum cost of 4 is a three-input, three-output reversible logic gate that combines the functionalities of the Toffoli gate and the Controlled NOT (CNOT) gate. It operates with inputs A,B and C producing three outputs as depicted in Fig. 2. The Peres gate is often utilized in quantum computing and low-power digital circuits due to its lower quantum cost compared to using separate Toffoli and CNOT gates.

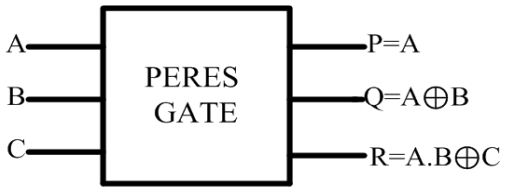


Fig. 2. Peres Gate

3)TSG Gate: The TSG (Toffoli, Swap, and NOT) gate is a reversible logic gate as depicted in Fig. 3. that combines the functions of the Toffoli gate, Swap gate, and NOT gate. It is a three-input, three-output gate designed to perform multiple logical operations simultaneously, enhancing the efficiency of reversible circuits. The TSG gate's ability to integrate multiple logical functions into a single reversible operation makes it a valuable tool in the design of advanced computational systems.The quantum cost of TSG gate is 17.

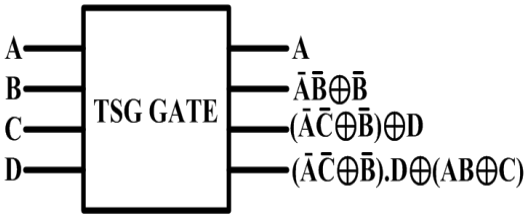


Fig. 3. TSG Gate

4)R2 Gate: The R2 gate is a 4x4 reversible gate, as depicted in Figure 4. This figure illustrates the R2 gate alongside its input and output mappings. The quantum cost of R2 gate is 5.

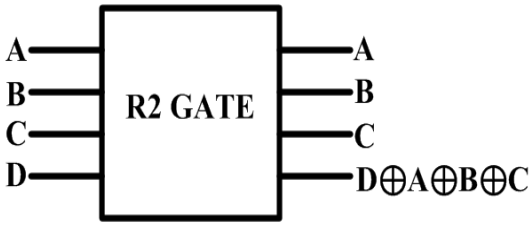


Fig. 4. R2 Gate

Reversible logic gates rely on specific properties to function within reversible computing systems. Here's a breakdown of some key parameters:

Quantum Cost(QC): Quantum cost refers to the total resources required to implement a reversible gate in a quantum computing system. This includes the number of basic quantum operations (such as quantum gates) needed, the depth of the circuit, and the complexity of the gates used.

Garbage Outputs(GO): Garbage outputs are the additional output lines produced by a reversible gate that do not directly contribute to the intended computation but are necessary to ensure the reversibility of the gate. Despite the fact that these outputs are not a part of the usable output, they are necessary in order to maintain the one-to-one mapping of input to output in reversible logic computing.

Gate Count(GC): This refers to the number of basic building block gates (either 1-input, 1-output or 2-input, 2-output) required to create the entire circuit.

Ancillae Inputs(AI): Ancillae (or Ancilla Inputs) refer to the additional input bits required in a reversible logic circuit to assist in the computation. These extra inputs are necessary for the construction of the reversible design, often used to facilitate certain operations while maintaining the reversibility of the circuit.

The paper's novel contributions include the utilization of various gates such as Peres, Feynman, TSG, and R2 in designing a 4-bit carry look-ahead adder. It introduces a new gate called the PGA gate, which employs Peres and Feynman gates and is used in the adder design. The proposed design outperforms existing architectures in terms of quantum cost.

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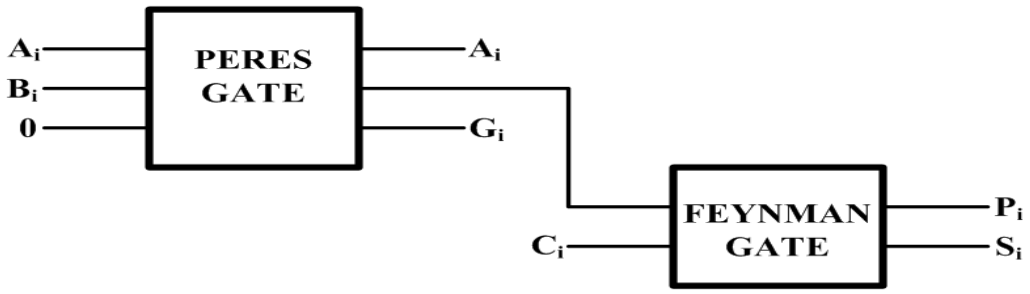


Fig. 5. Reversible PGA(Carry Propagate and Carry Generate Adder) Block

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Quantum cost is crucial in reversible logic gates as it measures the computational efficiency and resource utilization of quantum circuits. Lower quantum cost leads to more efficient and scalable quantum computing designs.

The organization of the paper is structured as follows: Section I offers an introduction to reversible logic gates and the Carry Look-Ahead (CLA) adder, highlighting their significance in contemporary computing. Section II provides a thorough literature review, exploring existing research on reversible gates and CLA adders, and identifying major advancements and current limitations. Section III outlines the proposed design for a CLA adder using reversible logic gates, describing the design methodology and logical framework. Section IV conducts a comparative analysis of the proposed design against existing literature. Lastly, Section V concludes the paper and suggests directions for future research.

II. Related Works

In the process of designing efficient adder units, the 4x4 reversible "TSG" gate was provided by H Thapliyal [16]. The demonstrated gate is employed in the creation of three separate types of carry adders, which are the ripple carry algorithm, the BCD algorithm, and the carry look-ahead algorithm. As one of the most important characteristics of the proposed gate, the capability of the suggested gate to carry out the functions of a reversible full adder is among the most important factors. Beginning from this point onward, it is now feasible to create a reversible full adder by making use of only a single gate. Adder designs that make use of the suggested gate have been shown to be significantly more efficient and optimised than its literature-based competing techniques. This has been proved through a number of studies. One way to demonstrate this is by establishing a correlation between the number of garbage outputs (GO) and the number of reversible gates installed. This article presents innovative recommendations for a reversible two-bit carry-lookahead adder that maintains nanometric parity. When compared to ideas that have been utilised in the past, Saeedeh Asri et al., [17] developed that the designs would be ideal in terms of the quantity of garbage outputs, constant inputs, and quantum costs. This was proven by comparing the designs to earlier conceptions. In addition, you should offer novel proposals for the nanometric parity-preserving reversible 4-bit carry-lookahead adder. This is for the aim of building circuits at a higher level. The researchers made the decision to proceed with their findings despite the fact that prior concepts that were presented as 4-bit did not exhibit the reversible parity-preserving quality.

S Sharmila Devi et al., [18] demonstrated the realisation of a Carry Look ahead Adder (CLA) by utilising a Reversible Partial Adder (RPA) and a Peres Gate (PG). This adder gains speed by utilising parallel carry computations. The output of the adder circuit is measured by comparing the newly presented research to the work that has been done in the past. It appears from the results of the simulation that the reversible CLA that was presented is capable of achieving superior reductions in area, power, and latency. It is possible to use the adder circuit as a beneficial element in the low power design for biomedical equipment due to the fact that it is both very durable and efficient in terms of energy consumption. Sarangam K et al., [19] proposed the method of opts for low-power architectures, including quantum computing and the Internet of Things. Typical combinational logic circuits are inefficient and consume energy. This becomes less of an issue with reversible computing, which preserves input bits in the output. This article presents a low-power reversible full adder that is optimised for energy-efficient applications. A greater comparison between energy-efficient full adders in reversible circuits and conventional CMOS circuits is recommended. Commonly used measurements for energy dissipation in complementary metal-oxide semiconductor (CMOS) circuits are $0.5CV_{dd}^2$, which is similar to $0.5CV_{Thr}^2$ in reversible logic. The amount of energy lost is equal to $(V_{Thr}/V_{dd})^2$. In addition, the size of ancillae and trash outputs has been reduced by 16.67%. There are also 16.67% fewer transistors than before.

III. Proposed Design

The Carry Look-Ahead (CLA) adder enhances arithmetic operation speed by calculating carry bits in parallel rather than sequentially. It achieves this through the generate (G_i) and propagate (Pi) functions for each bit position. The generate function (G_i=A_i . B_i) indicates when a carry is generated if both input bits are 1, while the propagate function (P_i=$A_i \oplus B_i$) shows when a carry is propagated if either input bit is 1 but not

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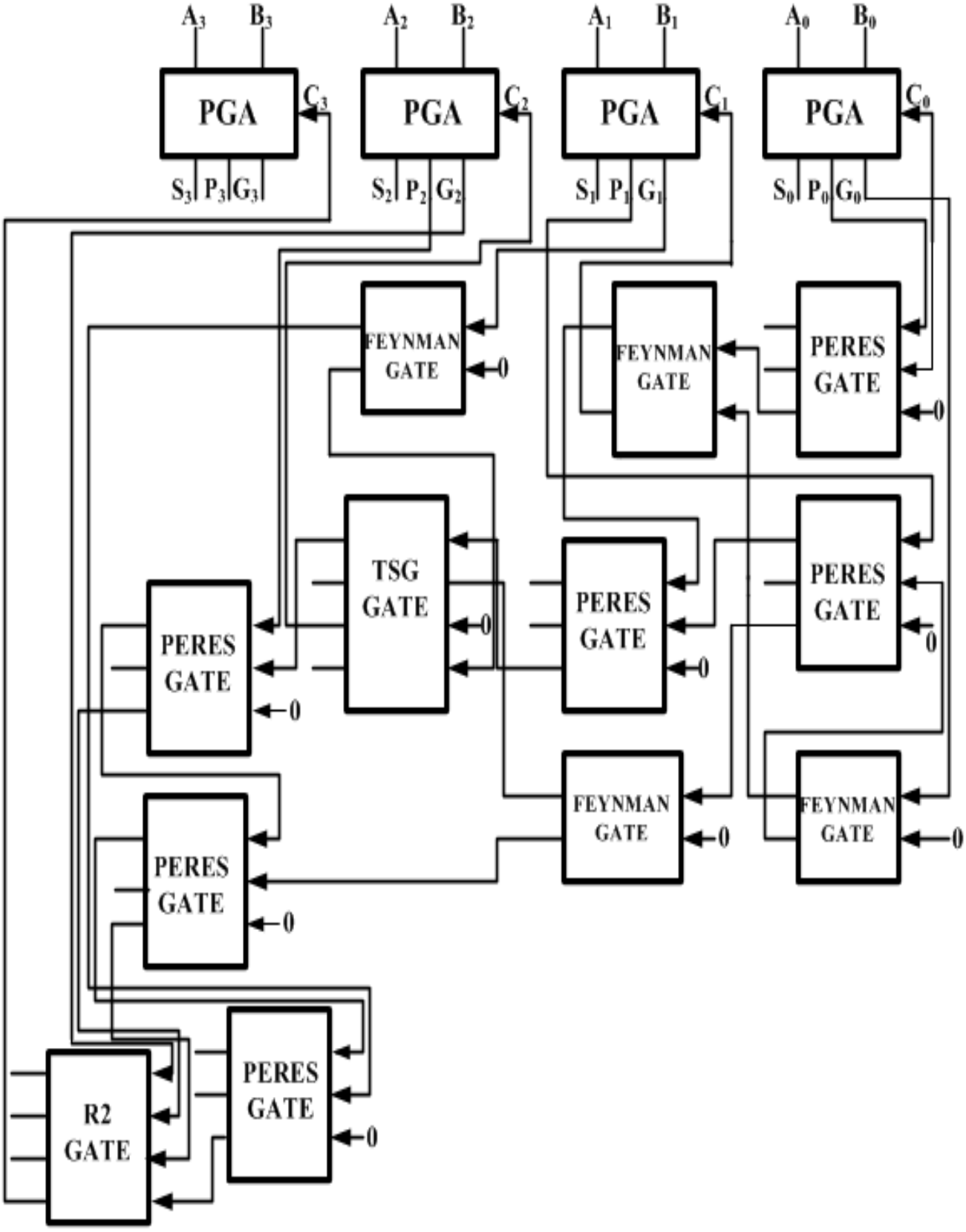


Fig. 6. Block diagram of Reversible 4-bit CLA Circuit

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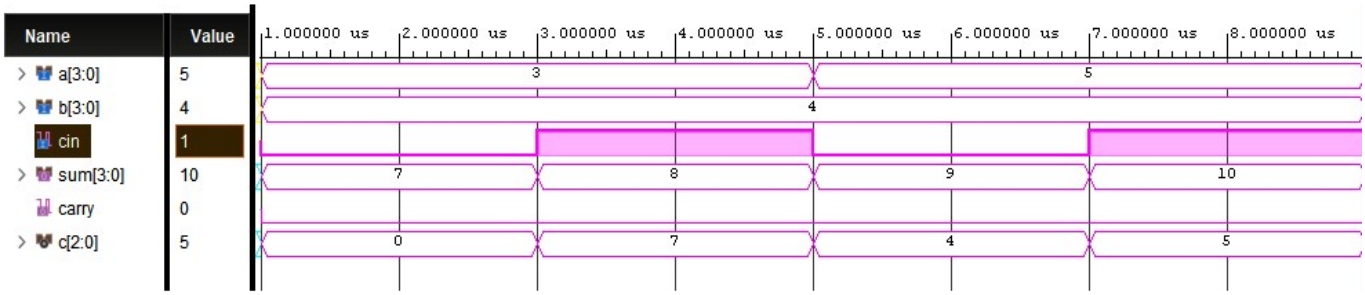


Fig. 7. Simulation Waveform of Reversible 4-bit CLA Circuit

TABLE I
COMPARISON OF PROPOSED 8 BIT ADDER/SUBTRACTOR WITH OTHER ARCHITECTURES

Design	Gate Count (GC)	Ancillae Inputs (AI)	Garbage Outputs (GO)	Quantum Cost (QC)	Total Reversible Logic Implementation Cost(TRLIC)
Existing Design[16]	19	16	27	139	201
Existing Design[17]	24	16	22	78	140
Proposed Design	20	14	14	70	118

both. Using these functions, the CLA adder computes carry outputs in parallel: the carry for each bit is determined by combining the generate and propagate values with the carry input from the previous bit position. For instance, C_1=G_0 + (P_0 . C_0) and C_2=G_1 + (P_1 . C_1) and so forth. Finally, the sum for each bit is calculated using the propagate function and the respective carry S_i = $P_i \oplus C_i$. This method of precomputing carry signals drastically reduces the delay caused by carry propagation, making CLA adders highly suitable for applications requiring rapid and efficient arithmetic computations.

The block that generates P_i, G_i and S_i is referred to as the PGA (Carry Propagate and Carry Generate Adder). The PGA block shown in Fig. 5.is built using a Peres gate with inputs A_i, B_i and 0. The third output of the Peres gate, G_i serves as the carry generate output. The second output of peres gate is connected to the first input of a Feynman gate, with the second input being C_i. This configuration produces two outputs: P_i(Carry propogate) and S_i(sum). The proposed 4-bit carry look ahead adder as depicted in Fig. 6. consists of PGA block, 4 feynman gates, 6 peres gates, TSG gate and R2 gate. In the design of reversible circuits, it is necessary to create copies of certain bits for computation, which introduces fan-out. However, fan-out is not allowed in reversible computation. This limitation requires a special type of gate for copying bits. When it comes to bit copying in reversible circuits, the Feynman gate reigns supreme. When we set the second input to zero, the first input gets mirrored at both outputs. This unique property allows the Feynman gate to create a clean copy of a bit without producing any garbage outputs.

IV. RESULTS AND DISCUSSIONS

The design of a novel gate utilizing reversible logic gate technology was evaluated using Vivado 2022.2 software, with the source code implemented in Verilog HDL. To synthesize and simulate Verilog code in Xilinx Vivado 2022.2, start by creating a new project and naming it appropriately while selecting the RTL Project type. Add your Verilog source files and set the top module for your design. Next, run synthesis by selecting “Run Synthesis” from the Flow Navigator, and review the results once the process is complete. To simulate, create a testbench for your design, add it to the project, and set it as the top module. Then, execute the behavioral simulation by choosing “Run Behavioral Simulation” from the Flow Navigator, and analyze the results in the waveform viewer.

The results of the designed 4-bit Carry Look-Ahead Adder were evaluated using various 4-bit input combinations for A and B. The performance was analyzed by testing all possible input scenarios, ensuring comprehensive validation. For example, with inputs A=0101 (5 in decimal) and B=0011 (3 in decimal), the CLA adder correctly computed the sum as 1000 (8 in decimal) and the carry-out. Similarly, with inputs A=1111 (15 in decimal) and B=0001 (1 in decimal), the sum was 0000(0 in decimal) with a carry-out of 1, indicating as overflow. Another test with inputs A=1010 (10 in decimal) and B=0101 (5 in decimal) resulted in a sum of 1111 (15 in decimal) and no carry-out. The parallel calculation of carry bits significantly reduced propagation delay compared to traditional adders, confirming the efficacy of the proposed design. The simulation wavefrom of the introduced circuitry is illustrated in Fig. 7.

The comparative analysis of the suggested architecture is presented in Table 1, which provides details concerning factors such as the number of gates, the ancilla inputs, the garbage outputs, and the quantum calculation. The architecture makes use of twenty gates, fourteen ancilla inputs, and fourteen garbage outputs, which results in a quantum cost of seventy and TRLIC of 118. The performance factor TRLIC outperfomes when compared to other related works.

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V. Conclusion and Future work

The proposed 4-bit Carry Look-Ahead Adder (CLA) designed using novel reversible logic gates demonstrates significant improvements in efficiency and performance. The design incorporates a robust configuration of Peres and Feynman gates to create the PGA block, facilitating efficient carry propagation and summation without generating additional garbage outputs. The novel gate design was tested using Vivado 2022.2 software, with Verilog HDL as the implementation language. The proposed design demonstrates a 41% improvement in TRLIC compared to related works. Overall, the proposed 4-bit CLA adder represents a substantial advancement in reversible logic design, offering a highly efficient and scalable solution for future computing technologies.

The proposed 4-bit Carry Look-Ahead Adder (CLA) that makes use of reversible logic gates has the capability of being scaled to larger bit widths, which enhances its use in quantum computing calculations that include more elaborate arithmetic operations. It is possible that future study could examine its application in a variety of quantum algorithms and complex computing models, taking advantage of its benefits in terms of speed and efficiency.

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